EX NO:1	Butterworth IIR Filters
DATE:	1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

```
Fs=10000; Fp=1000; Fst=1500;

[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);

[b,a]=butter(N,Wn);


freqz(b,a) % Magnitude & Phase


figure

[h,n]=impz(b,a,50);

...title('Impulse Response')


figure

stepz(b,a)


figure

zplane(b,a)


fprintf('Minimum Filter Order (N) = %d\n',N);

fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn); Input:
```

 Passband Ripple (Rp) = 1 dB

 Stopband Attenuation (Rs

) = 20 dB

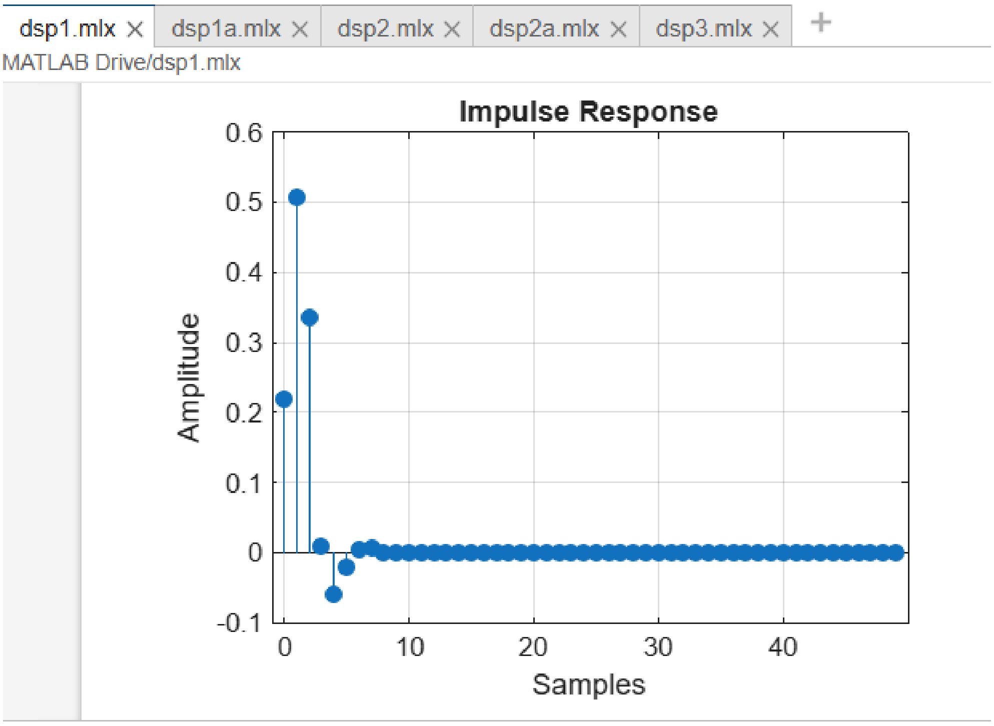
 Passband Frequency (Fp) = 1000 Hz

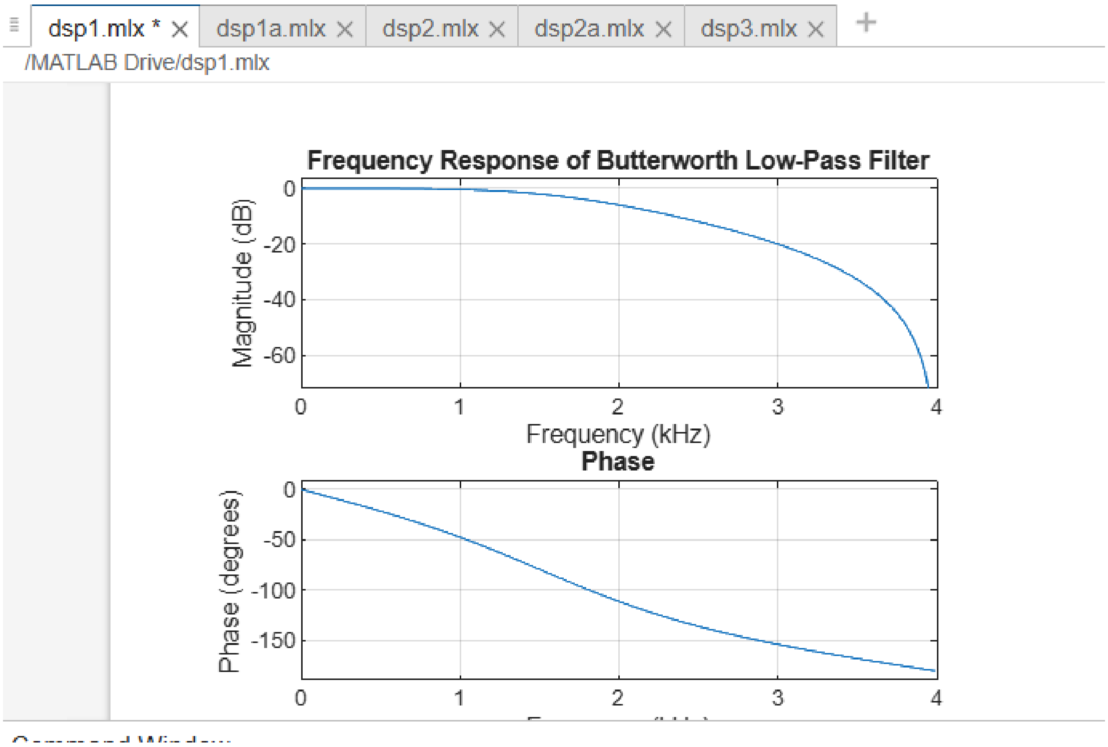
□ Stopband Frequency (Fs

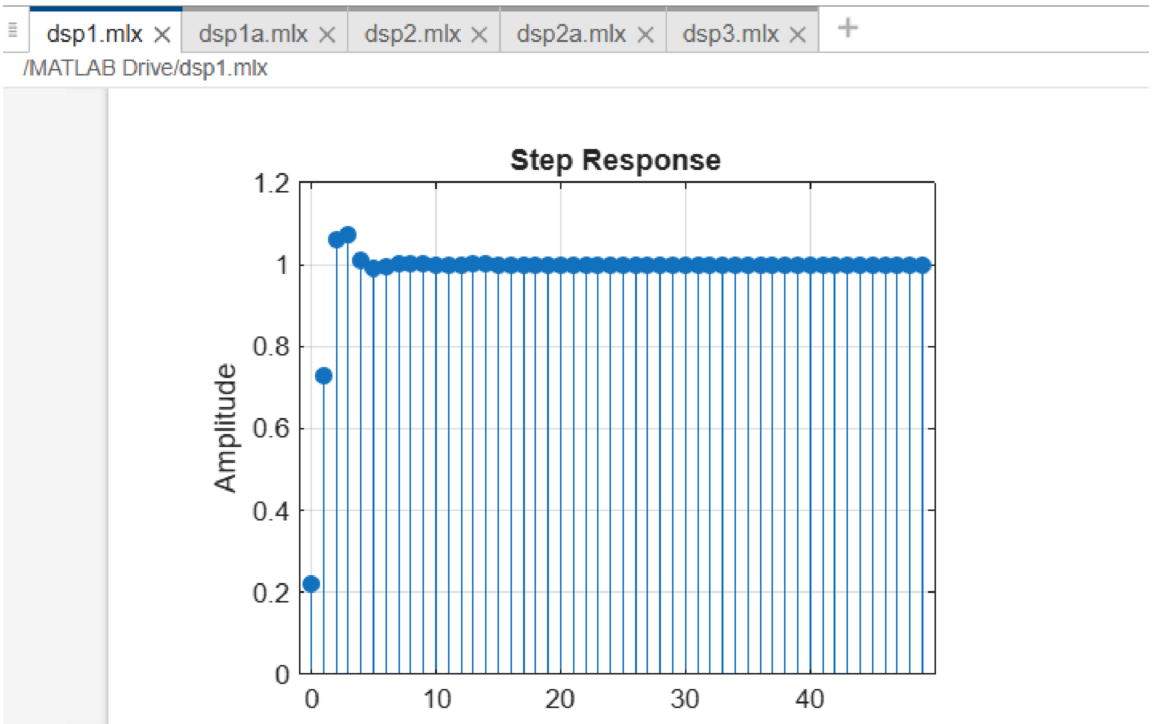
) = 3000 Hz

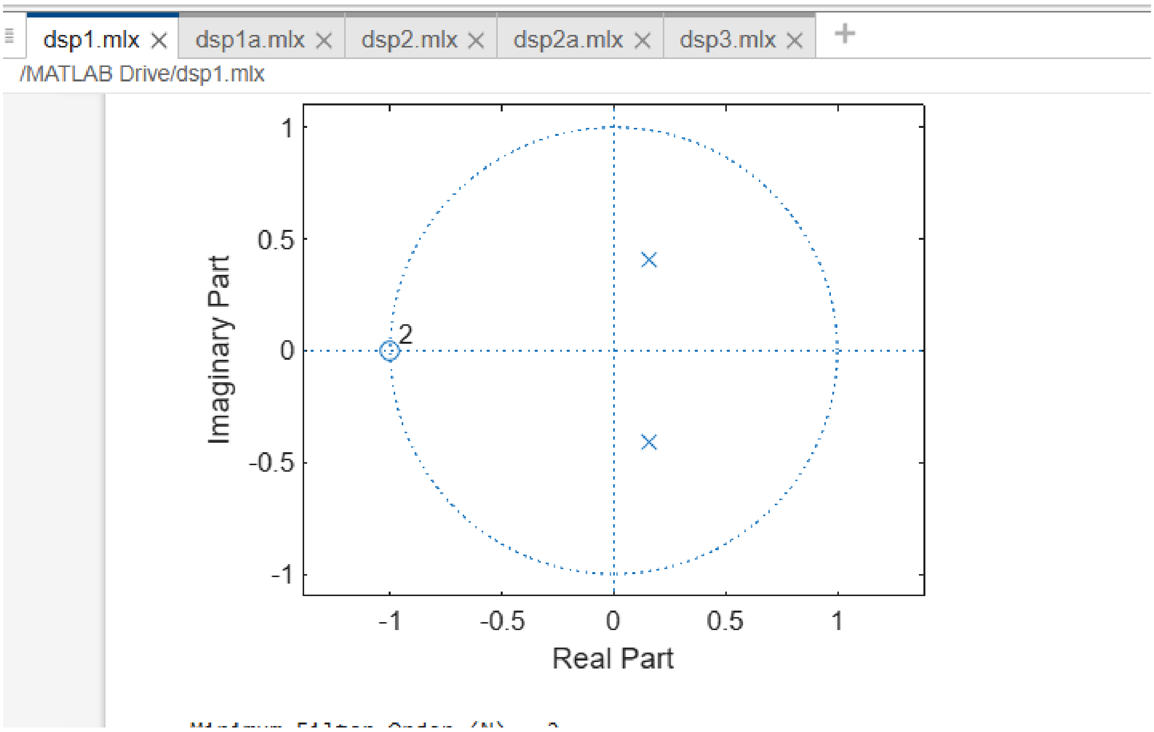
□ Sampling Frequency (Fsampling) = 8000 Hz

□ Filter Choice = Low-Pass Filter (LPF)




dsp1.mlx *
dsp1a.mlx
dsp2.mlx
dsp2a.mlx
dsp3.mlx
/MATLAB Drive/dsp1.mlx
Frequency Response of Butterworth Low-Pass Filter
Magnitude (dB)
0
-20
-40
-60
0 1 2 3 4
Frequency (kHz)
Phase
Phase (degrees)
0
-50
-100
-150
0 1 2 3 4

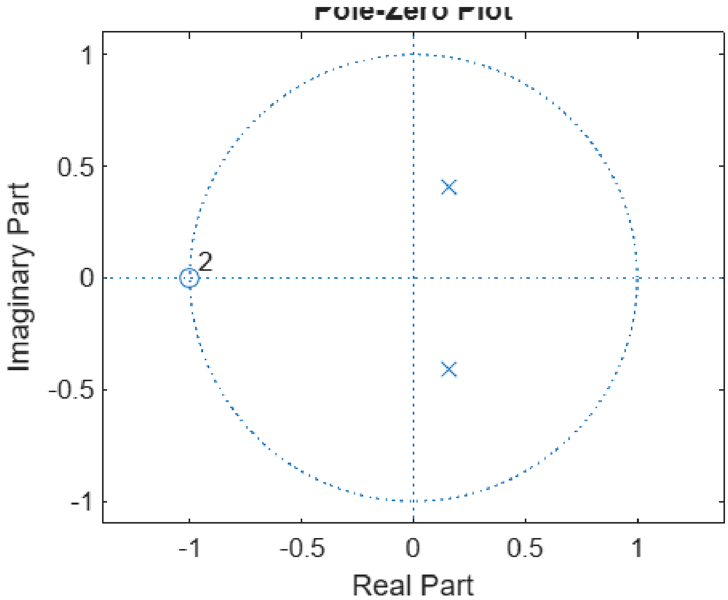

dsp1.mlx
dsp1a.mlx
dsp2.mlx
dsp2a.mlx
dsp3.mlx
/MATLAB Drive/dsp1.mlx
Step Response
Amplitude
1.2
1
0.8
0.6
0.4
0.2
0
0 10 20 30 40

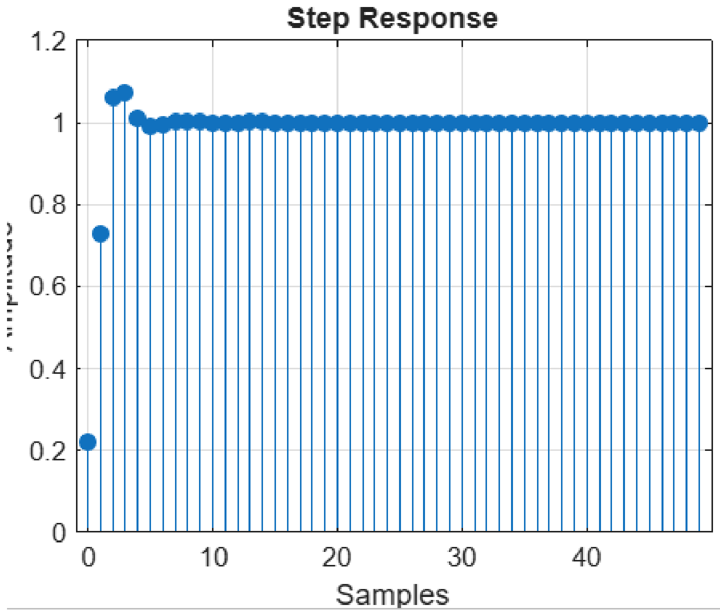


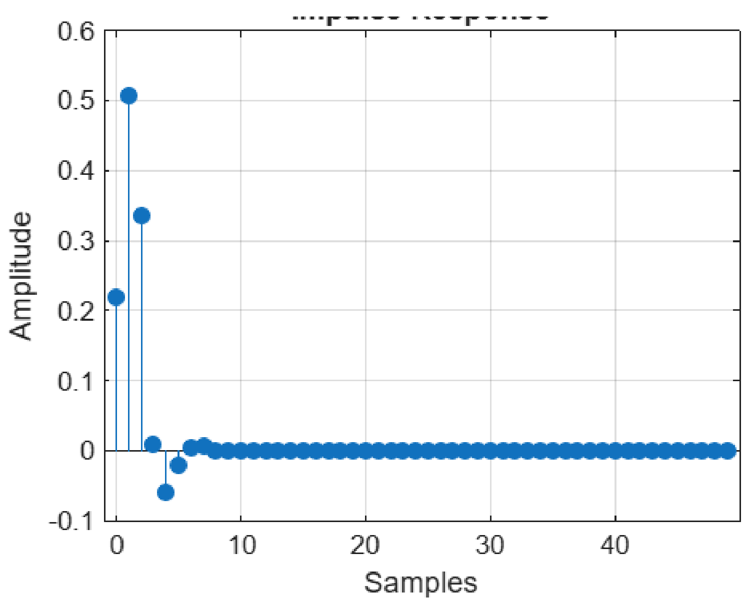
Ex no 1	Butterworth IIR Filters
date	1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

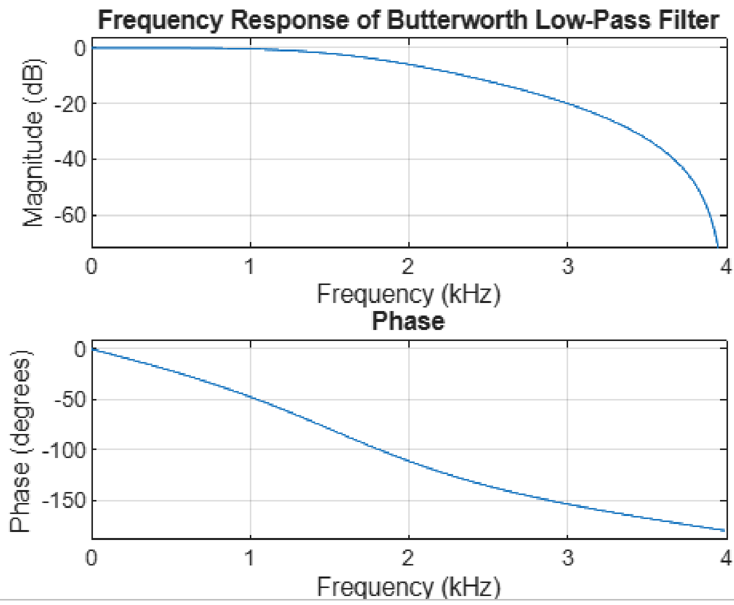
```
Fs=10000; Fp=3000; Fst=2000;

[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);

[b,a]=butter(N,Wn,'high');

% Magnitude & Phase

freqz(b,a)

% Impulse Response

figure

[h,n]=impz(b,a,50);
```

```
stem(n,h), grid on
title('Impulse Response')
% Step Response
figure
stepz(b,a)
title('Step Response')
% Group Delay
figure
grpdelay(b,a)
title('Group Delay')
% Pole-Zero Plot
figure
zplane(b,a)
title('Pole-Zero Plot')
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```


Pole-Zero Plot
Imaginary Part
Real Part
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
2


Step Response
1.2
1
0.8
0.6
0.4
0.2
0
0
10
20
30
40
Samples


Amplitude
0.6
0.5
0.4
0.3
0.2
0.1
0
-0.1
0
10
20
30
40
Samples


Frequency Response of Butterworth Low-Pass Filter
Magnitude (dB)
0
-20
-40
-60
0
1
2
3
4
Frequency (kHz)
Phase
Phase (degrees)
0
-50
-100
-150
0
1
2
3
4
Frequency (kHz)